

Hongwei Xi

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Education

Ph.D. in Pure & Applied Logic — *Carnegie Mellon University, USA, December 1998*
M.S. in Mathematics — *Nanjing University, China, July 1988*
B.S. in Mathematics — *Nanjing University, China, June 1985*

Academic Experience

Assistant Professor
09/99 to Present
Department of ECECS, University of Cincinnati
Teaching both undergraduate and graduate courses and supervising graduate students

Post-doc Research Associate
08/98 to 08/99
Department of Computer Science & Engineering
Oregon Graduate Institute

Doctoral Thesis
02/96 to 08/98
Dependent Types in Practical Programming
This work strengthens the current type system of the functional programming language ML with a restricted form of dependent types. It not only provides the programmer with significantly more information to catch program errors at compile-time, but also enables the compiler writer to obtain type information for optimizing code generation. As a significant application, we have shown how to eliminate array bound checks in realistic programs.

Instructor
Summer 1998
Carnegie Mellon Action Program
Calculus II for Humanities and Social Sciences

Instructor
Summer 1997
Carnegie Mellon Action Program
An introductory calculus course for minority students

Instructor
Summer 1997
Dept. of Mathematical Sciences, Carnegie Mellon Univ.
Calculus II for Humanities and Social Sciences

Instructor
Summer 1996
Dept. of Mathematical Sciences, Carnegie Mellon Univ.
Calculus II for Humanities and Social Sciences

Teaching Assistant
09/94 to 05/98
Dept. of Mathematical Sciences, Carnegie Mellon Univ.
Calculus I & II, Calculus in 3 dimensions, Modern Math.

Academic Experience

(continued)

Research Assistant
06/92 to 08/94 and Summer 95
Dept. of Mathematical Sciences, Carnegie Mellon Univ.
Maintaining and improving a very large theorem proving system TPS, which consists of more than 100,000 lines of code written in Common Lisp.

Assistant Professor
07/88 to 06/92
Dept. of Computer Science, Shanghai Jiao Tong University
Responsible for teaching *Mathematical Logic* and *Discrete Mathematics* at undergraduate level and *Type Theory* at graduate level. Also responsible for designing a programing language based upon equational reasoning and supervising the internship of undergraduate students.

Relevant Experience

Consultant
07/90 to 05/91
The 8th Construction Company, Shanghai
Designing a graphics system on PC for drawing flowcharts and monitoring construction schedules.

Consultant
06/91 to 06/92
The Construction Branch of Bao Shan Steel Factory, Shanghai
Using software monitoring the construction of the No. 3 furnace in Bao Shan Steel Factory, which was the largest in China at that time.

Selected Courses

All are graduate level ones taken at Carnegie Mellon University

Computer Science Programming languages, Computer architecture, Operating systems, Artificial intelligence, Algorithms

Mathematics Real analysis, Abstract algebra, Topology, Functional analysis, Mathematical Logic

Statistics Advanced probability theory I & II, Advanced statistics I & II

PhD Thesis

- Hongwei Xi, *Dependent Types in Practical Programming*, pp viii+189, December 1998.

Grant Awards

- Hongwei Xi (PI), *CAREER: Realistic Program Termination Verification: Theory and Practice*, National Science Foundation, CCR-0092703, $284,938, 2001-2006.
- Hongwei Xi (PI), *Imperative Programming with Dependent Types*, National Science Foundation, ITR-0081316, $334,918, 2000-2003.
- Paul Sivilotti and Hongwei Xi (PIs), *Dependent Types for High-Confidence Distributed Systems*, Ohio Board of Regents, $89,954, 2000-2002.
- Hongwei Xi (PI), *Faculty Development*, University of Cincinnati, $3,108, 2000.

Refereed Papers

Publication in Journals

- Femke van Raamsdonk, Paula Severi, Morten H. Sorensen and Hongwei Xi, *Perpetual Reductions in lambda-calculus*, Journal of Information and Computation 149(2):173-225, March 1999.
- Hongwei Xi, *Upper Bounds for Standardization and an Application*, Journal of Symbolic Logic, 64(1), pp. 291-303, March 1999. An earlier version with the same title appeared in Proceedings of the 5th Kurt Gödel Colloquium, 1997.
- Peter B. Andrews, Matthew Bishop, Sunil Issar, Dan Nesmith, Frank Pfenning, and Hongwei Xi, *TPS: A Theorem Proving System for Classical Type Theory*, Journal of Automated Reasoning, 16(3), pp. 321-353, 1996.
- Hongwei Xi, *On recursively enumerable degrees* (in Chinese with English abstract), Math. Semi-annual of Nanjing University, China, 1989.

Publication in Conferences

- Hongwei Xi, *Dependent Types for Program Termination Verification.* In Proceedings of 16th IEEE Symposium on Logic in Computer Science (**LICS '01**), Boston (to appear), June 2001.
- Hongwei Xi, *Imperative Programming with Dependent Types.* In Proceedings of 15th IEEE Symposium on Logic in Computer Science (**LICS '00**), pp. 375-387, Santa Barbara, June 2000.
- Hongwei Xi and Songtao Xia, *Towards Array Bound Check Elimination in Java Virtual Machine Language.* In Proceedings of CASCON '99, pp. 110-125, Mississauga, Ontario, November 1999.
- Hongwei Xi and Frank Pfenning, *Dependent Types in Practical Programming.* In Proceedings of ACM SIGPLAN Symposium on Principles of Programming Languages (**POPL '99**), pp. 214–227, San Antonio, January 1999.
- Hongwei Xi and Frank Pfenning, *Eliminating Array Bound Checking Through Dependent Types.* In Proceedings of ACM SIGPLAN Conference on Programming Language Design and Implementation (**PLDI '98**), pp. 249–257, Montreal, June 1998.
- Hongwei Xi, *Towards Automated Termination Proofs through "Freezing".* In Proceedings of 9th International Conference on Rewriting Techniques and Applications (RTA '98), LNCS, vol. 1379, pp. 271–285, Tsukuba, April 1998.
- Hongwei Xi, *Evaluation under Lambda-Abstraction.* In Proceedings of the 9th International Symposium on Programming Languages, Implementations, Logics, and Programs (PLILP '97), LNCS, vol. 1292, pp. 259-273, Southampton, September 1997.
- Hongwei Xi, *Upper Bounds for Standardization and an Application.* In Proceedings of the 5th Kurt Gödel Colloquium (KGC '97), LNCS, vol. 1289, pp. 335-348, Vienna, August 1997. A detailed version has appeared in the Journal of Symbolic Logic.
- Hongwei Xi, *Simulating Eta-Expansions with Beta-Reductions in the Second-Order Polymorphic Lambda-Calculus.* In Proceedings of Logical Foundations of Computer Science (LFCS '97), LNCS, vol. 1234, pp. 399-409, Yaroslavl, July 1997.
- Hongwei Xi, *Weak and Strong Beta Normalizations in Typed Lambda-Calculi.* In Proceedings of Typed Lambda Calculi and Applications (TLCA '97), LNCS, vol. 1210, pp. 390-404, Nancy, April 1997.

Publication in Workshops

- Hongwei Xi, *Dependently Typed Data Structures.* In Proceedings of Workshop on Algorithmic Aspects of Advanced Programming Languages (WAAAPL '99), Paris, September 1999.
- Hongwei Xi, *Dead Code Elimination through Dependent Types.* In Proceedings of First International Workshop on Practical Aspects of Declarative Languages (PADL '99), LNCS vol. 1551, pp. 228–242, San Antonio, January 1999.

Refereed Papers *(continued)*

Publication in Workshops *(continued)*

- Hongwei Xi, *Generalized Lambda-Calculi*(abstract). In the meeting report of the 4th Workshop on Logic, Language, Information and Computation (WoLLIC '97), Logic Journal of Interest Group in Pure and Applied Logics (IGPL), vol. 5(6), pp. 925-927, Fortaleza (Ceará), August 1997.
- Peter B. Andrews, Matthew Bishop, Sunil Issar, Dan Nesmith, Frank Pfenning, and Hongwei Xi, *TPS: An interactive and automatic tool for proving theorems of type theory*, In Jeffrey J. Joyce and Carl-Johan H. Seger, editors, Proceedings of the 6th International Workshop on Higher-Order Logic Theorem Proving and Its Applications, *Springer-Verlag LNCS 780*, pp 366-370, Vancouver, August 1993.
- Hongwei Xi, *On branching and nonbranching recursively enumerable degrees* (in Chinese), National Logic Conference, Shantou University, China, October 1990.

Unrefereed Papers

Technical Reports

- Hongwei Xi and Joachim Steinbach, *Erasure for Termination Proofs*, Technical report OGI-CSE-99-009, Oregon Graduate Institute, August 1999.
- Hongwei Xi and Robert Harper, *Dependently Typed Assembly Languages*. Technical report OGI-CSE-99-008, Oregon Graduate Institute, July 1999.
- Joachim Steinbach and Hongwei Xi, *Freezing – Termination Proofs for Classical, Context-Sensitive and Innermost Rewriting*, Institut für Informatik, Technische Universitat Muenchen, January 1998.
- Hongwei Xi, *On Weak and Strong Normalizations*, Research Report 96-187, Mathematics Department, Carnegie Mellon University, 1996.
- Hongwei Xi, *An Induction Measure on Lambda-Terms and Its Applications*, Research Report 96-192, Department of Mathematical Sciences, 1996.

Manuscripts

- Hongwei Xi, *Dependent Types for Robust Programs*, August, 2000.
- Hongwei Xi, *Combining Algebraic Rewriting with Second Order Extensional Polymorphic Lambda Calculus*, October, 1996.
- Hongwei Xi, *Separating Developments*, October, 1996.

Technical Writing

- Dan Nesmith, Matthew Bishop, Peter Andrews, Sunil Issar, Frank Pfenning, and Hongwei Xi, *TPS User's Manual*, pp. 84+iii.
- Peter Andrews, Sunil Issar, Dan Nesmith, Frank Pfenning, Hongwei Xi and Matthew Bishop, *TPS3 Facilities Guide for Programmers and Users*, pp. 238+viii.
- Frank Pfenning, Sunil Issar, Dan Nesmith, Peter Andrews, Hongwei Xi and Matthew Bishop, *ETPS User's Manual*, Version for Basic Logic, Mathematical Logic I & II, pp. 60+ii.

Invited Talks

- Hongwei Xi, *Compiling with Dependent Types*, Invited talk at proof-carrying code (PCC) workshop, Santa Barbara, June 2000.
- Hongwei Xi, *A Dependently Typed Assembly Language*, ROPAS group, Department of Computer Science, Korea Advanced Institute of Sicence and Technology (KAIST), Taejon, Korea, 7 April, 1999.
- Hongwei Xi, *Dependent Types in Practical Programming* ROPAS group, Department of Computer Science, Korea Advanced Institute of Sicence and Technology (KAIST), Taejon, Korea, 6 April, 1999.
- Hongwei Xi, *A Dependently Typed Assembly Language*, Invited talk at Workshop on Dependent Types in Programming, Applied Semantics (APPSEM) group, Göteborg, Sweden, 27 - 28 March, 1999.
- Hongwei Xi, *Dependent Types in Practical Programming*, Invited talk at Formal Methods PI meeting, Stanford University, 29 - 30 October, 1998.

Conference Presentations

- Hongwei Xi, *Imperative Programming with Dependent Types*, 15th IEEE Symposium on Logic in Computer Science (LICS '00), Santa Barbara, June 2000.
- Hongwei Xi, *Dependent Types in Practical Programming*, Symposium on Principles of Programming Languages (POPL '99), San Antonio, January, 1999.
- Hongwei Xi, *Dead Code Elimination through Dependent Types*, International Workshop on Practical Aspects of Declarative Languages (PADL '99), San Antonio, January, 1999.
- Hongwei Xi, *Eliminating Array Bound Checking Through Dependent Types*, Programming Language Design and Implementation (PLDI '98), Montreal, June, 1998.
- Matthew Bishop and Hongwei Xi, *The ETPS Educational Theorem Proving System*, The 8th Annual Computing and Philosophy Conference (CAP), Pittsburgh, August, 1993.

Graduate Student Supervision

- Songtao Xia, PhD student, Oregon Graduate Institute, August 1998 - August 1999.
- Varun Nayak, MS student, University of Cincinnati, January 2000 - Present.
- Zaiwei Du, PhD student, University of Cincinnati, September 2000 - Present.

Other Information

Citizenship People's Republic of China
Visa Type H-1B
Referee CSER (Consortium for Software Engineering Research) Book (2000); The First International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT) (2000); IEEE Symposium on Logic in Computer Science (LICS) (2000); Journal of Higher-Order Symbolic Computation (2000, 1999); International Conference on Functional Programming (ICFP) (1998); Journal of Information and Computation (1996)
Membership Association of Computing Machinery (ACM); American Mathematical Society (AMS); Kurt Gödel Society (KGS)
Languages Fluent in Chinese and English.

References

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Russell Walker (teaching reference)
Senior Lecturer and Associate Head
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